

# Ali Ahmadi Esfidi

AI Researcher & Developer

✉ aliahmadiesfidi@outlook.com | 🌐 Mr-Ahmadi | 📄 Ali Ahmadi Esfidi | 🌐 Portfolio

## Education

**B.Sc. Computer Science** > Amirkabir University of Technology 📅 Sep 2022 – Sep 2026

- Location: Tehran, Iran • Field of study: Computer Science • GPA: 18.93/20 • Top 20%

**High School Diploma** > Dr. Shahriari High School (SAMPAD) 📅 Sep 2019 – Jun 2022

- Location: Qom, Iran • Field of study: Mathematics & Physics • GPA: 18.42/20

## Research Interests

- Deep Reinforcement Learning
- Machine Learning for Decision-Making Systems
- AI for Infrastructure Monitoring
- Optimization Methods for AI and Networking
- Computational Biology & Bioinformatics
- Software-Defined Networks & Resource Allocation

## Experiences

**Research Assistant** > Supervisor: Prof. Ghathe 📅 Jan 2025 – Present

📍 NORC Lab, Amirkabir University of Technology

- **Led** an AI team and **engineered** web systems and deep learning models for automated bridge damage detection in collaboration with the Tehran Urban Planning and Research Center.
- **Gained hands-on experience** in developing efficient deep-learning pipelines for computer vision applications, focusing on model compactness and structural health monitoring.
- **Stack:** PyTorch • OpenCV • ReactJS • NodeJS • PostgreSQL • Docker

**Research Assistant** > Supervisor: Prof. Khonsari 📅 Sep 2024 – Present

📍 High Performance Networks Lab, Tehran University

- **Developed** deep reinforcement learning systems for complex agricultural decision-making, focusing on reward engineering and policy optimization to balance multiple objectives.
- **Acquired proficiency** in benchmarking reinforcement learning algorithms across discrete and continuous action spaces, and in utilizing simulation environments for training and evaluation.
- **Stack:** PyTorch • TensorFlow • Pandas • NumPy

**Research Assistant** > Supervisor: Dr. Dolati 📅 Sep 2025 – Present

📍 Sharif University of Technology

- **Formulated** MDPs for switch pairing and collaborative rule caching, and designed simulation environments for joint scheduling-routing policies in programmable data planes.
- **Implemented** optimization frameworks for in-network aggregation in distributed machine learning, focusing on ILP decomposition, flexible routing strategies, and scalable algorithmic design.
- **Stack:** PyTorch • Numba • Pandas • NumPy

**Teaching Assistant** > Graduate & Undergraduate Courses 📅 Sep 2024 – Present

📍 Amirkabir University of Technology & Tehran University

|                                                                    |                               |                     |
|--------------------------------------------------------------------|-------------------------------|---------------------|
| • <b>Data Mining</b> <span>B.Sc.</span>                            | Dr. Yousefimehr               | Sep 2026 – Present  |
| • <b>Quantum Information Processing</b> <span>M.Sc.</span>         | Prof. Khonsari                | Sep 2025 – Jan 2026 |
| • <b>Computational Data Mining</b> <span>M.Sc.</span>              | Prof. Ghathe                  | Sep 2025 – Jan 2026 |
| • <b>Introduction to Logic</b> <span>B.Sc.</span>                  | Dr. Didehvar                  | Sep 2025 – Jan 2026 |
| • <b>Theory of Computation</b> <span>B.Sc.</span>                  | Dr. Didehvar                  | Sep 2024 – Jun 2026 |
| • <b>Artificial Intelligence &amp; Workshop</b> <span>B.Sc.</span> | Prof. Ghathe, Dr. Yousefimehr | Sep 2024 – Jan 2025 |

## Publications

- M. Saberi, A. Ahmadi Esfidi, M. Dolati, N. Omidvar, **FlexINA: In-Network Aggregation for Accelerating Distributed Machine Learning with Flexible Routing**, Published in *IEEE Transactions on Network and Service Management*. [View Paper](#).

- P. Heidari, A. Khonsari, [A. Ahmadi Esfidi](#), A. Mehrvarz, E. Khodaei, A. Dadlani, **DeepIrrigo: Deep Reinforcement Learning for Continuous and Discrete Action Irrigation Optimization — A Case Study in Iran**, Under review at *Intelligent Systems with Applications*. [View Paper](#).
- M. Saberi, [A. Ahmadi Esfidi](#), M. Dolati, A. Khonsari, A. Movaghar, **Rule Caching in Programmable Networks with Deep Reinforcement Learning**, Submitted to *IEEE Transactions on Network and Service Management*.
- P. Heidari, [A. Ahmadi Esfidi](#), A. Mehrvarz, E. Khodaei, A. Khonsari, **Irrigation Optimization in Agricultural Fields Using Deep Reinforcement Learning Approaches**, Published in *CSICC 2025*. [View Paper](#).

## Selected Projects

 [See more](#)

### DRL-Based Beamforming Optimization for Movable Intelligent Surfaces

 Jun 2026 – Present

- **Designed** a DRL framework to optimize beamforming for MIS-assisted SWIPT systems, jointly configuring phase shifts and power splitting ratios to maximize energy harvesting.
- *Stack:* [PyTorch](#) • [Gymnasium](#) • [MOSEK](#) • [NumPy](#)

### RL-Based ML Job Scheduler with Distribution Optimization

 Aug 2025 – Oct 2025

- **Developed** a two-stage hierarchical PPO system for scheduling ML jobs on multi-server accelerator clusters using the Gavel dataset, reducing average job completion time.
- *Stack:* [PyTorch](#) • [Gymnasium](#) • [NumPy](#)

### Machine Learning Perspectives on Gold-Binding Peptides

 May 2025 – Jun 2025

- **Applied** machine learning to classify and predict gold-binding affinity of peptide sequences, comparing regression and classification approaches across multiple featurization strategies.
- *Stack:* [PyTorch](#) • [Scikit-learn](#) • [Hugging Face](#) • [Pandas](#) • [NumPy/SciPy](#)

### RNA Pairing Pattern Recognition Model

 Mar 2024 – Jul 2024

- **Developed** a system to predict RNA secondary structures using SCFGs, evolutionary information, and enhanced CYK parsing for complex motifs; presented at [CBRC Journal Club](#).
- *Stack:* [NetworkX](#) • [Biopython](#) • [PhyML](#) • [NumPy/SciPy](#)

## Skills

|                       |                                                                                                                                                                                           |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Programming Languages | <a href="#">Python</a> • <a href="#">JavaScript</a> • <a href="#">TypeScript</a> • <a href="#">R</a> • <a href="#">Java</a> • <a href="#">C++</a> • <a href="#">Swift</a>                 |
| Frameworks            | <a href="#">React</a> • <a href="#">Node.js</a> • <a href="#">Express.js</a> • <a href="#">Django</a> • <a href="#">Electron</a>                                                          |
| Data Science / ML     | <a href="#">PyTorch</a> • <a href="#">TensorFlow</a> • <a href="#">Pandas</a> • <a href="#">Scikit-learn</a> • <a href="#">NumPy</a> • <a href="#">OpenCV</a> • <a href="#">LangChain</a> |
| Platforms & Tools     | <a href="#">Docker</a> • <a href="#">Git</a> • <a href="#">GitHub</a> • <a href="#">MongoDB</a> • <a href="#">PostgreSQL</a> • <a href="#">Hugging Face</a>                               |

## Certifications

### Bioinformatics Internship Program > [Biocan](#)

 Nov 2025

Scientific Secretary: Dr. Z. Salehi • Course Director: Dr. K. Kavousi

### Introduction to Bioinformatics > [Biocan](#)

 Jul 2025

Scientific Chair: Dr. K. Kavousi • Score: 96/100 • Top 3% of 350+ students

### Scrum Foundations Course > [Ultima Training Tech](#)

 Dec 2024

Instructor: Josef Balahan

## Volunteering

### Student Scientific Magazine “Halghe” > Amirkabir University of Technology

 Dec 2025 – Present

Contributed to scientific editing and supported the editorial team

### Technical & Web Support Volunteer > Amirkabir University of Technology

 May 2023, Feb 2025

Web and procurement support for DAI DAY (2025) and the International Conference on AI and Smart Vehicles (2023)